

Research and Development Report

AI/ML Internship — Day 20

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Project Folder: Raritone

Introduction

During the AI/ML Internship, most of my assigned work was related to Artificial Intelligence, Computer Vision, Virtual Try-On systems, body measurement estimation, and recommendation technologies. While working on these individual modules, I gradually realized that a complete AI product cannot be developed by a single team.

The Raritone platform represents a collaborative environment where Artificial Intelligence, Backend Development, and Data Analytics work together to create a seamless and intelligent fashion ecosystem. Every team performs specialized tasks, but the overall success of the platform depends on continuous interaction between these departments.

This research summarizes my understanding of the complete project after studying the responsibilities and workflows of different technical teams involved in the development process.

Understanding the Project Vision

The primary objective of the project is to improve the online fashion shopping experience through Artificial Intelligence.

Traditional e-commerce platforms allow customers to browse products and place orders, but they often fail to solve one major problem: customers cannot physically try garments before purchasing them.

The proposed platform addresses this challenge by integrating intelligent technologies such as:

- AI-powered Virtual Try-On.
- Personalized fashion recommendations.
- Body measurement estimation.
- Smart product analysis.
- Business analytics and trend prediction.

The platform combines these technologies to reduce product returns, increase customer confidence, and improve overall user satisfaction.

Overall Structure of the Platform

The platform consists of several interconnected modules.

The major participants are:

Customers

Customers browse products, upload images, receive AI-generated try-on results, and obtain personalized outfit recommendations.

Vendors

Vendors upload product information, garment images, pricing details, and inventory data.

Backend Services

Backend systems manage the communication between users, databases, AI modules, and business operations.

Artificial Intelligence Engine

The AI engine processes images, estimates body measurements, performs garment fitting, and generates recommendations.

Data Analytics Module

The analytics system studies customer interactions and business data to improve future decision-making.

Although these modules perform different functions, they work together as a single integrated system.

My Understanding of the AI/ML Team

As an AI/ML intern, this was the department I explored in the greatest detail.

The Artificial Intelligence team is responsible for developing features that make the platform intelligent and interactive.

The major responsibilities include:

Human Detection

The AI system first identifies whether a valid human body is present in the uploaded image.

Body Segmentation

The system separates the user from the background and identifies important body regions.

This improves the quality of garment placement.

Pose Estimation

Pose estimation detects body joints and landmarks that help AI understand the orientation of the user.

Models researched:

- MediaPipe Pose
 - OpenPose
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Body Measurement Extraction

The AI system estimates measurements such as:

- Height
- Shoulder Width
- Waist Width
- Hip Width

These measurements help improve garment fitting accuracy.

Virtual Try-On

The AI engine performs:

- Garment segmentation.
- Garment warping.
- Cloth alignment.
- Image blending.

This generates the final try-on image.

Recommendation System

AI also analyses user preferences and previous interactions to recommend suitable clothing items.

The recommendation approaches studied include:

- Content-Based Filtering.
 - Collaborative Filtering.
 - Hybrid Recommendation Models.
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My Understanding of the Backend Team

The Backend Team acts as the operational foundation of the entire platform.

While AI generates intelligent outputs, backend systems ensure that all business operations run smoothly.

Their responsibilities include:

- User account management.
- Authentication and authorization.
- Vendor registration.
- Product management.
- Inventory updates.
- Order processing.
- Payment integration.
- API development.
- Database management.

The backend team also ensures secure communication between the AI modules and customer-facing applications.

Without backend services, even the most advanced AI models cannot function effectively.

My Understanding of the Data Analytics Team

The Data Analytics Team transforms large amounts of platform data into useful business knowledge.

Every interaction within the platform generates valuable information.

Examples include:

- Product views.
- Customer searches.
- Purchase history.
- Return statistics.
- Vendor performance.
- Inventory movement.

The analytics team studies these patterns to help the company make better business decisions.

The insights generated can also be used to improve AI recommendation systems and demand forecasting.

Interaction Between Different Teams

One of the most interesting observations during this internship was understanding how different teams depend on one another.

When a vendor uploads a product, the backend stores the information.

The AI team processes garment images and prepares them for Virtual Try-On.

Customers interact with these AI services.

Their interactions generate additional business data.

The analytics team studies this data and identifies patterns that help improve the overall platform.

The improved insights are then used by both the backend and AI teams to optimize future operations.

Thus, the platform continuously evolves through collaboration.

Shared Technologies Across Teams

Although every department has unique responsibilities, several technologies are shared across the project.

Technology	Application
Python	AI and Analytics
SQL	Database Operations
GitHub	Version Control
REST APIs	Module Communication
Cloud Storage	Data Management
Docker	Deployment
AWS	Cloud Infrastructure

The use of common technologies enables efficient collaboration among teams.

Professional Learning Experience

This research changed my understanding of software development.

Initially, I believed that Machine Learning models alone could create intelligent applications.

However, the internship demonstrated that successful AI products require the combined efforts of multiple engineering domains.

Artificial Intelligence provides intelligence.

Backend systems provide stability.

Data Analytics provides business understanding.

Together, they create a complete technology solution.

Importance of Cross-Functional Collaboration

Modern software projects are rarely built by isolated teams.

Cross-functional collaboration improves:

- Product quality.
- Development speed.
- Business decision-making.
- User experience.
- Scalability.

The Raritone project is a good example of how specialized technical teams can work together toward a common objective.

Research Outcome

This study provided a broader understanding of how large-scale AI products are designed and managed.

Beyond learning Machine Learning concepts, I gained insight into software architecture, backend operations, data management, and interdisciplinary collaboration.

The experience strengthened my understanding of the relationship between Artificial Intelligence, Backend Engineering, and Data Analytics.

Conclusion

Studying the overall project helped me understand that building an AI-powered fashion platform requires much more than developing Machine Learning models. The platform depends on the coordinated efforts of several specialized teams that work together to achieve common goals.

The AI/ML Team develops intelligent features that improve personalization and customer experience. The Backend Team manages business operations, databases, and system communication. The Data Analytics Team transforms operational data into valuable business insights that guide future improvements.

This cross-functional approach creates a scalable, efficient, and intelligent ecosystem capable of supporting modern digital fashion applications. The research expanded my technical perspective and demonstrated the importance of collaboration in developing real-world AI solutions.
